A Two-Grid Characterization of Euler Sampling Accuracy in Flow Matching

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***Abstract*—A flow-matching model learns a velocity field that moves noise toward a data distribution. More solver steps can follow this field more closely yet produce a worse match to the data. This can happen because numerical error offsets an error in the field. We ask what accuracy at two step counts reveals about the field. We run explicit Euler with the same field from Gaussian noise on a grid of N steps and on the grid that halves each step. At the shared times, we compare both outputs with the target distributions using the quadratic Wasserstein distance. Under common derivative bounds, the sum of these errors can decrease faster than $1/N$ exactly when a limit of the allowed fields generates the target path and its acceleration preserves each distribution. Second-order accuracy is then possible. Otherwise the error sum is at least a positive constant times $1/N$ on all sufficiently fine grids. For the path that joins independent noise and data by straight lines, this lower bound holds in odd dimension for centered targets with positive definite covariance. For a scalar Gaussian path on uniform grids, the sharp constant is $(2+\pi)/8$. For each of 20 scalar neural fields fitted at small step counts, going from 16 to 64 steps reduces the integration error and increases the error against the target. This difference matters when a sampler is used at a different step count from the one used in training.**

***Impact Statement*—Flow-matching models use numerical steps to generate samples. With explicit Euler, each step evaluates the learned network once, so changing the step count changes the cost of sampling. We show why a model that works well at one step count can give worse samples at another. More steps bring the solver closer to the exact motion, but a coarse solver can offset some of the error in the field itself. Our theory gives conditions under which the output can stay accurate at two step counts. In the experiments, we also find that a small evaluation set can reverse the apparent direction of an improvement. When comparing samplers, report the error against the target and the integration error separately. Give the step counts, the times used for comparison, and the evaluation set.**

***Index Terms*—Flow matching, generative models, numerical integration, optimal transport, ordinary differential equations, sampling efficiency.**

I. Introduction

Flow matching trains a neural network to give a *velocity field* [1–3]. The field tells a particle how to move at each time and position. An ordinary differential equation (ODE) describes this motion, and the motion carries random noise to a data distribution. Diffusion models have an equivalent deterministic sampler [4], and large image models use this method [5].[1]

A sampler solves the ODE with a numerical method in a finite number of steps. Each step uses at least one evaluation of the network, so the step count sets the cost of each sample. Much work decreases this count with better solvers [10–13], with straighter trajectories [14–16], and with models that are trained for few steps [17–20]. Each method tunes a field or a solver for some step counts.

Two different errors occur. The *integration error* is the distance between the numerical run and the exact motion of the learned field. The *target error* is the distance between the numerical distribution and the intended distribution. On fine grids, Euler approaches the exact motion. The target error can still increase if the coarse run offsets part of the field's error.

Fig. 1(a) gives an example with a one-dimensional Gaussian (normal) distribution. The standard deviation is 1 at the start, and the target standard deviation at time $1/2$ is approximately 0.71. Section VII gives a field for which one Euler step to time $1/2$ gives the target distribution exactly. Two steps of length $1/4$ with the same field give a standard deviation of approximately 0.85, and the exact motion of the field gives 1.02. The one large step offsets an error of the field. A smaller step removes the offset.

Prior work explains the forward direction of this effect. Different velocity fields can generate the same distributions at every time [21]. Take a field that generates the target path. Only the part of its acceleration that changes the distribution controls the leading Euler error. A field without that part permits a higher order [22, 23]. We ask the converse question for a field that can be biased. If the same field is accurate with N and $2N$ Euler steps, what does that accuracy show about the field?

The answer is a property of the continuous motion. In each comparison we use one field on both grids. Across comparisons, the field can change with the step count if all fields satisfy the same derivative bounds. Suppose that the two target errors decay faster than $1/N$. Then a limit of the allowed fields generates the full target path, and its acceleration preserves each distribution. If no such limit exists, the sum of the two target errors has a first-order lower bound. Our contributions are as follows.

- *Two-grid characterization.* Theorem 1 gives the equivalence and the uniform lower bound c/N. It separates three regimes of the best two-grid error: a positive limit, first order, and second order.
- *Bound at finite step counts.* Theorem 2 uses the two target errors to bound the part of the acceleration that changes the distributions of the field. We identify the grid refinements and the pairs of solvers that permit this inference.
- *Independent interpolation path.* A common path joins independent noise and data by straight lines. For this path,

[1] AI assistance for this section is disclosed in the acknowledgment [6–9].

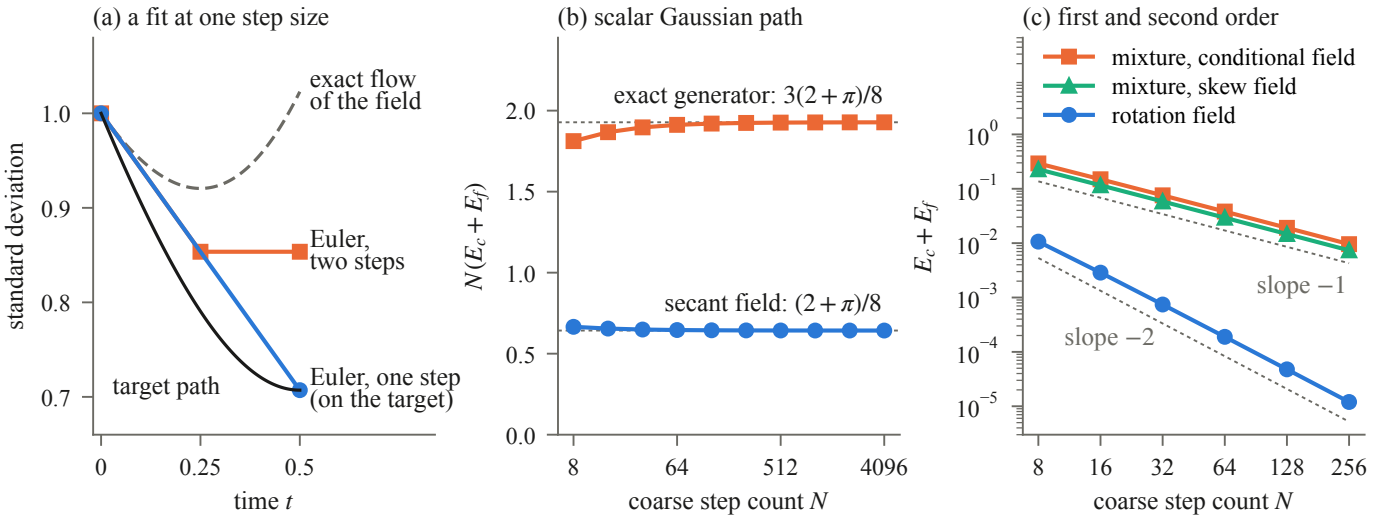


Fig. 1. Exact population computations. (a) A field that is fitted to one Euler step of size $1/2$ gives the target standard deviation at time $1/2$. Two steps of size $1/4$ with the same field do not. (b) For the scalar Gaussian path of Section VII, the scaled two-grid error $N(E_c + E_f)$ converges to $(2+\pi)/8$ for the secant field and to three times that value for the exact generator. (c) The two-grid error decays at first order for two exact generators of a three-dimensional mixture path, and at second order for a rotation field whose acceleration preserves each distribution.

Theorem 3 proves the first-order lower bound in every odd dimension and for all targets with positive definite covariance. Theorem 4 gives a necessary condition on the regression loss at time zero in every dimension.

- *Sharpness.* A scalar Gaussian path has the sharp constant $(2+\pi)/8$. Examples show that the smoothness assumptions and the use of intermediate times are necessary.
- *Experiments.* Exact population computations agree with the predicted rates. Learned scalar fields show, in the population Wasserstein distance, how a fit at small step counts trades numerical error against model bias. A study of 230 evaluation sets measures how sample size affects a comparison of two step counts.

II. Problem and Regularity Class

A probability path $(p_t)_{0\le t\le T}$ gives the target distribution at each time. We measure the distance between two distributions with the quadratic Wasserstein distance W_2 [24]. Pair the points of the two distributions and take the root mean squared distance between paired points. The smallest value over all pairings is W_2. A joint distribution of paired points is a *coupling*. If numerical and exact particles start from the same point, they give one coupling, and its error is an upper bound on W_2.[2]

Fix a final time $0 < T \le 1$, a dimension $d \ge 1$, and a derivative bound $K \ge 1$. Let p_t be a W_2-continuous path of distributions with finite second moment that starts at the standard Gaussian $p_0 = \gamma_d = \mathcal{N}(0, I_d)$. We write $\mathbb{E}$ for expectation and $|x|$ for the Euclidean norm. Let $\mathcal{B}_K$ be the class of twice continuously differentiable (C^2) velocity fields u on a neighborhood of $[0,T]\times\mathbb{R}^d$ that satisfy

$$\begin{aligned} |u(t,x)|,\ |\partial_t u(t,x)| &\le K(1+|x|),\\ |\partial_{tt} u(t,x)| &\le K(1+|x|)^2,\\ \|D_x u\|,\ \|D_x\partial_t u\|,\ \|D_x^2 u\| &\le K. \end{aligned} \tag{1}$$

[2]AI assistance for this section is disclosed in the acknowledgment [6–9].

Here D_x is the derivative in space, ∂_t is the derivative in time, and $\|\cdot\|$ is the operator norm. The bounds hold at every point of $[0,T]\times\mathbb{R}^d$. They limit how quickly the velocity can change in space and in time. They are not bounds on the size of a network.

A sequence of fields in this class can converge to a field outside the class. We write $\overline{\mathcal{B}}_K$ for the closure of $\mathcal{B}_K$ in C^1_{loc}, in which the fields and their first derivatives converge uniformly on each compact set. The closure is compact. Each sequence has a subsequence that converges in this sense. The first derivatives of a limit field are locally Lipschitz. On each compact set, their change is at most a constant times the distance between the inputs.

Let $F^u_{s,t}(x)$ be the position at time t of the particle that is at x at time s and that solves $\dot X_t = u(t, X_t)$. The field has its own distributions $\rho^u_t = (F^u_{0,t})_{\#}\gamma_d$, which are the distributions of $X^u_t = F^u_{0,t}(G)$ for $G \sim \gamma_d$. A fitted field need not satisfy $\rho^u_t = p_t$. The *model bias* is $\mathfrak{b}(u) = \sup_t W_2(\rho^u_t, p_t)$. Let $\mathrm{Law}(X)$ be the distribution of a random variable X. On a grid $\mathcal{T} = (0 = t_0 < \cdots < t_N = T)$ with steps $h_k = t_{k+1} - t_k$ and $H = \max_k h_k$, explicit Euler computes

$$\widehat X_{k+1} = \widehat X_k + h_k u(t_k, \widehat X_k), \qquad \widehat X_0 \sim \gamma_d.$$

Divide each interval into two equal parts and run Euler again with the same field and the same initial distribution. Let $\widetilde X_{2k}$ be the state of this refined run at the coarse time t_k. The coarse error, the refined error, and the *two-grid error* are

$$\begin{aligned} E_c(u,\mathcal{T}) &= \max_k W_2(\mathrm{Law}(\widehat X_k), p_{t_k}),\\ E_f(u,\mathcal{T}) &= \max_k W_2(\mathrm{Law}(\widetilde X_{2k}), p_{t_k}),\\ \mathcal{J}_N(u,\mathcal{T}) &= E_c + E_f. \end{aligned} \tag{2}$$

We measure both errors at the times of the coarse grid. The two runs use N and $2N$ evaluations of the field.

An *exact generator* is a field with $\rho_t^u = p_t$ for all t. The *acceleration* of a field is $a_t(x) = \partial_t u(t,x) + D_x u(t,x) u(t,x)$. It is the rate of change of the velocity of a moving particle.

Definition 1 (Order-elevating generator)**.** A field $u \in \overline{\mathcal{B}}_K$ is an order-elevating generator of p if $\rho_t^u = p_t$ for all t and its acceleration satisfies

$$\int \nabla\phi(x) \cdot a_t(x)\, p_t(\,\mathrm{d}x) = 0 \tag{3}$$

for all smooth functions ϕ with compact support and all $t \in [0,T]$.

To read (3), hold the time fixed and move the particles for a short interval in the direction a_t. The integral is the rate of change of the average of the test function ϕ. If the integral is zero for all tests, every test average has zero initial rate of change. Under our regularity assumptions, the motion with time fixed preserves p_t. Particles can still move. We then say that the acceleration *preserves the distribution*. For a smooth density the condition is $\nabla \cdot (p_t a_t) = 0$ [25].

III. Two-Grid Characterization

The first result answers the main question. The two target errors can decay faster than first order exactly when the closure contains an order-elevating generator. The result is about the best fields that the common bounds permit, not about each field in the class.[3]

Theorem 1 (Two-grid characterization)**.** *For the target path and the class of Section II, these statements are equivalent.*

1) *There are step counts $N_j \to \infty$, fields $u_j \in \mathcal{B}_K$, and N_j-step grids $\mathcal{T}_j$ with $H_j \to 0$ such that $\mathcal{J}_{N_j}(u_j, \mathcal{T}_j) = o(N_j^{-1})$.*
2) *The closure $\overline{\mathcal{B}}_K$ contains an order-elevating generator of p.*
3) *On the uniform grids $t_k = kT/N$, there are fields $u_N \in \mathcal{B}_K$ with $\mathcal{J}_N(u_N, \mathcal{T}_N) = O(N^{-2})$.*

If the closure contains no order-elevating generator, there are constants $c, H_0 > 0$ such that, for every $u \in \mathcal{B}_K$ and every deterministic N-step grid with $H \le H_0$,

$$E_c(u,\mathcal{T}) + E_f(u,\mathcal{T}) \ge c/N. \tag{4}$$

The field and the grid can change with N. Each pair of a coarse run and a refined run must use one field.

The notation $o(N^{-1})$ means that N times the error tends to zero, and $O(N^{-2})$ means that the error is at most a constant times N^{-2} for large N. The theorem is a statement about the limit $N \to \infty$. It does not give an error rate from one experiment. The maximum of the two errors can replace their sum. The constant in the lower bound then changes by a factor of at most two. The implication from statement 2 to statement 3 is the direction of the prior results [22, Theorem 2], [23, Corollary 8], here for limit fields in $\mathbb{R}^d$. The other implications and the lower bound are new.

[3]AI assistance for this section is disclosed in the acknowledgment [6–9].

A. Proof Idea: Subtract the Two Runs

Euler uses the velocity at the start of each step. It therefore omits the displacement $h^2 a_t/2$ that the acceleration causes. To see the effect on a distribution, compare the averages of one scalar test in the two runs. Let $\phi_t = \phi \circ F_{t,0}^u$ carry a test ϕ backward along the flow of the field, so that its value is constant along each exact trajectory. Let $I_u(t) = \mathbb{E}[\nabla\phi_t(X_t^u) \cdot a_t(X_t^u)]$ measure how the acceleration changes this average. Each Euler step changes the average by $-h_k^2 I_u(t_k)/2$ to leading order, and these changes add [26]:

$$\mathbb{E}\phi_{t_j}(\widehat{X}_j) - \mathbb{E}\phi(G) = -\tfrac{1}{2}\sum_{k<j} h_k^2 I_u(t_k) + O\Big(H \sum_{k<j} h_k^2\Big).$$

The constants are the same for all fields in $\mathcal{B}_K$ and all tests with common derivative bounds. In the refined run each step is divided in two, so the two squared half steps sum to half the squared coarse step. Subtract the refined expansion from the coarse expansion, as in step doubling for the estimation of ODE errors [27, 28]:

$$Z_j - Z_i = -\tfrac{1}{4}\sum_{k=i}^{j-1} h_k^2 I_u(t_k) + O\Big(H \sum_{k=i}^{j-1} h_k^2\Big). \tag{5}$$

Here Z_k is the difference of the two numerical averages at t_k. Both runs have the same exact-flow average, so the unknown model bias cancels. The target distribution is a common reference for the two runs, and the triangle inequality gives $|Z_k| \le C(E_c + E_f)$. We do not need the distance between ρ_t^u and p_t.

Suppose that the two-grid error is $o(N^{-1})$ and $H \to 0$. Compactness gives a subsequence of fields with a limit. The integration error of Euler is $O(H)$ uniformly in the class, so the model bias of these fields tends to zero, and the limit generates the target path exactly. Suppose that (3) fails for the limit at some time and for some test. The flow maps are invertible, so this test is ϕ_t for a suitable ϕ, and I_u is not zero at that time. Then I_u keeps one sign on a time interval, also for the fields near the limit. On a fine grid, the first and last nodes in that interval are separated by at least some $\ell > 0$. The squared steps between them then sum to at least ℓ^2/N. Equation (5) then gives a difference of order $1/N$, which contradicts the assumed accuracy. Thus (3) holds.

Conversely, suppose that an exact generator satisfies (3). Move the particles along the acceleration field for an auxiliary time $h^2/2$ while the physical time is fixed. This motion changes the pairing of start points but not their distribution. The new pairing supplies the displacement that Euler omits, and the one-step W_2 error becomes $O(h^3)$. Stability of the Euler map gives a total error $O(H^2)$. If the generator is only a limit field, smooth fields in the class approximate both runs. Supplementary Sections S1–S3 give the uniform bounds and the full proof.

B. Three Regimes of the Best Two-Grid Error

On uniform grids let $q_N = \inf_{u \in \mathcal{B}_K} \mathcal{J}_N(u)$ and $\mathfrak{b}_* = \min_{u \in \overline{\mathcal{B}}_K} \mathfrak{b}(u)$. Three cases occur, and they depend on what

the closure contains:

$$\begin{array}{ll} \text{no exact generator:} & q_N \to 2\mathfrak{b}_* > 0, \\ \text{exact, not order-elevating:} & q_N = \Theta(N^{-1}), \\ \text{an order-elevating generator:} & q_N = O(N^{-2}). \end{array} \tag{6}$$

Here $\Theta(N^{-1})$ means upper and lower bounds of that order. The three cases separate model bias from numerical error, and they describe the best fields of the class. The last case includes exact Euler sampling. For one fixed field, both target errors approach its model bias as the step count increases.

IV. A Bound at Finite Step Counts

Theorem 1 is a limit statement. Two runs at finite step counts answer a related question, also when the field is biased. How much can the acceleration change the distributions of the field?[4]

In a test average, the field is paired with a gradient. Let $L^2(\rho)$ be the space of fields with finite mean squared norm under ρ. Let Π_ρ retain the part of a field detected by gradient tests. It is the orthogonal projection in $L^2(\rho)$ onto the closure of the gradients of smooth functions with compact support [25]. The *projected-acceleration residual* of a field is

$$\varepsilon_t(u) = \|\Pi_{\rho_t^u} a_t\|_{L^2(\rho_t^u)}.$$

A zero residual means that the acceleration preserves ρ_t^u. The agreement of ρ_t^u with p_t is a separate requirement.

The bound permits general refinements. Divide the coarse interval k into parts with fractions $\theta_{kr} > 0$, $\sum_r \theta_{kr} = 1$, and let

$$\beta_k = 1 - \sum_r \theta_{kr}^2 \geq \beta_0 > 0. \tag{7}$$

The number β_k is the fraction by which the sum of squared steps decreases in interval k. A refinement is *balanced* if a fixed $\beta_0 > 0$ is a lower bound for all β_k. Division into m equal parts has $\beta_0 = 1 - 1/m$, and bisection has $\beta_0 = 1/2$. Let E_r be the target error of the refined run at the coarse nodes, and let $E = E_c + E_r$. For bisection, $E_r = E_f$.

Theorem 2 (Finite-step bound on the residual)**.** *Let $u \in \overline{\mathcal{B}}_K$, and let the refinement satisfy (7). Let $\eta = 1/3$ for arbitrary grids and $\eta = 1/2$ for uniform grids. For every integer $q \geq 0$,*

$$\sup_t \varepsilon_t(u)^2 \leq e^{2KT} \left\{ \sum_{1 \leq |\alpha| \leq q+1} \left[L_\alpha H + \Lambda_\alpha (NE)^\eta \right]^2 + \frac{V}{q+1} \right\}. \tag{8}$$

The sum is over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_d)$ of total degree $|\alpha| = \sum_i \alpha_i$. The constants L_α, Λ_α, and V are explicit functions of K, d, T, β_0, and α. Supplementary Section S7 gives them, with $V = dM_2V_1^2$ in its notation. The bound does not assume that u generates the target path.

The bound has two parts. The sum controls finitely many coefficients of the acceleration in a basis of Gaussian (Hermite) polynomials [29]. Each coefficient is a test average. A large average keeps its sign on a time interval. The subtraction in (5) then gives a lower bound on the two target errors. The term $V/(q+1)$ controls the coefficients of higher degree, which the tests do not measure. It uses a bound on the derivative of the acceleration in the coordinates of the Gaussian start. Obzherin and Khrulkov [23, Proposition 10] separate measured coefficients from an orthogonal remainder. Here the two runs bound the measured coefficients, and regularity bounds the remainder.

The bound has this use in the limit. First choose q large to make the last term small. Then make H and NE small. The full residual then tends to zero uniformly in time. The constants increase quickly with the degree, the dimension, and K, so the bound is too large for numerical use in high dimension. A linear bound from tests with bounded derivatives would be stronger, but it is false. For $u_n(t,x) = t\sin(nx)/n^2$, which is in $\mathcal{B}_1$, the residual at time zero is asymptotic to $1/(\sqrt{2}n^2)$, while all such test averages are $O(n^{-3})$.

A. Informative Refinements and Solver Pairs

A refinement must change the step sizes by a sufficient amount. Theorem 1 and its lower bound hold for each balanced refinement, with $\beta_0 > 0$ common to all comparisons. A larger step count alone is not sufficient. For example, divide each step $h = 1/N$ into h/N and $h(1-1/N)$. Then $\beta_k = O(1/N)$, and both target errors can be $O(N^{-2})$ on a path that has no order-elevating generator. The step counts N and $N+1$ can be equally uninformative.

The same method compares two solvers on one grid. Consider a fixed explicit Runge–Kutta method. Its weights b sum to one and each stage time in c equals the sum of its row of stage coefficients and lies in $[0,1]$. The method has the coefficient $\kappa = 1/2 - \mathsf{b}^\top c$ in its leading particle error $-\kappa h^2 a$ [28]. If two methods have different values of κ, their leading errors remain after subtraction, and the characterization holds with their two target errors (Supplementary Section S8). Euler and Heun are one such pair. Two second-order methods both have $\kappa = 0$, so they are not such a pair.

V. The Independent Interpolation Path

We now apply the characterization to a common path. Draw noise and a data point independently, and interpolate between them:

$$p_t = \mathrm{Law}((1-t)G + tY), \qquad G \sim \gamma_d, \quad G \perp Y. \tag{9}$$

Here $G \perp Y$ means that G and Y are independent. The target Y has mean zero, a finite second moment, and a positive definite covariance S. Restrict the path to an interval $[0,T]$.[5]

Theorem 3 (Odd-dimensional obstruction)**.** *For odd d, the path (9) has no order-elevating generator in $\overline{\mathcal{B}}_K$. Thus there are constants $c, H_0 > 0$ such that (4) holds for every field in $\mathcal{B}_K$ and every N-step grid with $H \leq H_0$. The fields need not generate the path.*

The theorem is not limited to linear fields, Gaussian targets, radial targets, or one fixed field. For isotropic Gaussian paths

[4] AI assistance for this section is disclosed in the acknowledgment [6–9].

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and linear fields, the effect of the parity of the dimension is known [22, Theorem 12]. Theorem 3 extends it to nonlinear fields and to targets that are not Gaussian.

A. Why the Start of the Path Decides

At time zero a particle at x is paired with independent data of mean zero, so its average interpolation velocity is $-x$. For an exact limit generator, let $r(x) = u(0,x) + x$ be the difference from this velocity. The first time derivative of the path gives $\nabla \cdot (\gamma_d r) = 0$, so r does not change the Gaussian start. The second time derivative gives a second condition. For each unit vector e and each smooth compactly supported scalar test ψ,

$$\mathbb{E}[\psi'(e \cdot G)\, e \cdot a_0(G)] = \int \psi''(y)\{e^\top S e - Q_e(y)\}\varphi(y)\, \mathrm{d}y. \tag{10}$$

Here φ is the standard one-dimensional Gaussian density, and $Q_e(y) = \mathbb{E}[(e \cdot r(G))^2 \mid e \cdot G = y]$ is the mean squared correction in the direction e. If the acceleration preserves the distribution, the left side is zero for all ψ. We therefore look for an interval of y in which $Q_e(y)$ is smaller than the target variance $e^\top S e$.

For odd $d \geq 3$, a sphere in $\mathbb{R}^d$ has even dimension. The hairy-ball theorem says that a continuous field tangent to such a sphere is zero at some point [30]. Thus the tangential part of r is zero at some point Re of each sphere of radius R. Near this point, the condition $\nabla \cdot (\gamma_d r) = 0$ and the Lipschitz constant $\mathrm{Lip}(r)$ give $Q_e(y) = O(\mathrm{Lip}(r)^2/R^2)$, while $e^\top S e \geq \lambda_{\min}(S) > 0$. For large R, a convex test with ψ'' supported in $[R, R+1]$ makes the right side of (10) positive. This test has no compact support, and a cutoff argument extends (3) to it. The positive value therefore contradicts (3). The case $d = 1$ is direct. Supplementary Section S4 gives the proof. The same topological fact gives a related one-step result in [23, Theorem 34].

B. Exact Generators Exist, and Even Dimension Is Different

The obstruction can hold even when an exact generator exists. For $Y = \mu\xi e_1 + \sigma Z$, with an equally likely sign $\xi \in \{-1, 1\}$ and an independent standard Gaussian Z, the conditional velocity is a linear field plus a term $c(t)\tanh(b(t)x_1)e_1$. It generates the path and it satisfies (1) for a finite K.

Odd dimension is necessary. For even d and $Y \sim \mathcal{N}(0, \sigma^2 I)$, take a matrix J with $J^\top = -J$ and $J^2 = -I$, and let $X_t = ((1-t)I + \sigma t J)G$. This process has the distributions of (9), its trajectories are straight lines, and Euler is exact on every grid. Its field is the skew field of [22, Theorem 12] for this path. The process uses a different coupling of noise and data but the same path of distributions.

The obstruction also holds for a class of starts that are not Gaussian. Supplementary Theorem S9 proves it in odd dimension for radial start densities $q_W(x) \propto \exp\{-|x|^2/2 - W(|x|^2)\}$ with $W \in C^2([0,\infty))$, $W(0) = 0$, and $W' \geq 0$. Theorem 1 extends to positive continuous start densities with a finite fourth moment. The Hermite bound of Theorem 2 needs the Gaussian start.

VI. A Condition on the Initial Regression Loss

Theorem 3 uses only the target. In every dimension, the regression loss at time zero gives a second test for faster than first-order accuracy. On the path (9), flow matching predicts the displacement $Y - G$. Given $G = x$, the best prediction is $-x$ because Y has mean zero. Let $\nu = \operatorname{tr} S = \mathbb{E}|Y|^2$. The initial loss and its excess over the minimum are[6]

$$\begin{aligned} \mathcal{L}_0(u) &= \mathbb{E}|u(0,G) - (Y-G)|^2 = \nu + \Delta_0(u), \\ \Delta_0(u) &= \mathbb{E}|u(0,G) + G|^2. \end{aligned} \tag{11}$$

For an exact generator whose acceleration preserves each distribution, $\Delta_0 = \nu$, so the initial loss is twice its minimum. This follows from the moment bound of [23, Proposition 30] and the decomposition of the loss [31]. The next result does not assume exact generation. It uses only the errors of the two Euler runs.

Theorem 4 (Initial-loss inequality)**.** *For the path (9) in any dimension, where S can be singular, every $u \in \mathcal{B}_K$ and every deterministic N-step grid satisfy*

$$\begin{aligned} |\Delta_0(u) - \nu| \leq \omega_K(E_c + C_{\mathrm{path}}H) + C_H H \\ + \{64 D_q N(E_c + E_f)/c_\ell^2\}^{1/3}. \end{aligned} \tag{12}$$

On the uniform grid $h = T/N$, the last term can be replaced by

$$\{32 D_q N(E_c + E_f)/(T c_\ell)\}^{1/2}. \tag{13}$$

The constants C_{path}, C_H, D_q, c_ℓ and the function ω_K are explicit in K, d, T, ν (Supplementary Section S5). The function ω_K is nondecreasing, with $\omega_K(0) = 0$ and $\omega_K(b) = O(b^{1/3})$ as $b \to 0$. Thus, for fields $u_N \in \mathcal{B}_K$ on N-step grids with $H_N \to 0$, the accuracy $E_c + E_f = o(N^{-1})$ implies

$$\Delta_0(u_N) \to \nu, \qquad \mathcal{L}_0(u_N) \to 2\nu. \tag{14}$$

A fixed gap $|\Delta_0 - \nu| \geq g > 0$ gives explicit constants $c_g, H_g > 0$ such that $E_c + E_f \geq c_g/N$ on every grid with $H \leq H_g$.

The proof measures the acceleration at time zero with the quadratic test $|x|^2/2$ and uses the subtraction of the two runs. It estimates the first two time derivatives of the second-moment bias by finite differences. The function ω_K is the result of the best choice of the difference interval.

Three limits of this result are important. First, the condition is on the full expected loss at time zero. An average of the training loss over sampled times does not establish it. Second, the condition is only necessary: a three-dimensional skew field has $\Delta_0 = \nu$ and an acceleration that does not preserve the distribution (Supplementary Section S6). A larger regression loss is not a general method to improve a sampler. Third, the constants are conservative. For $d = T = K = 1$ and $\nu = g = 1$, the explicit threshold H_g is approximately 1.3×10^{-14}, which needs approximately 7.7×10^{13} uniform steps.

[6] AI assistance for this section is disclosed in the acknowledgment [6–9].

VII. Sharpness

This section gives the example of Fig. 1(a) in general form. A field can be exact at one step count and lose this accuracy after refinement. Let $p_t = \mathcal{N}(0, g(t)^2 I_d)$ with $g(t) = \sqrt{(1-t)^2 + \sigma^2 t^2}$ and $\sigma > 0$. For the uniform step $h = 1/N$, the *secant field* is[7]

$$u_h(t,x) = \frac{g(t+h) - g(t)}{h\, g(t)} x. \tag{15}$$

Its coefficient uses the change of the target scale over one full step. On its own grid, each Euler step multiplies each particle by $g(t+h)/g(t)$. Euler therefore gives each target distribution at the nodes exactly, in every dimension. These fields have common derivative bounds. The exact flow of u_h has a bias $\Theta(h)$. If the *same* field is run with the step $h/2$, the scale of the refined run is

$$\widehat{g}_{h/2;u_h}(t) = g(t) \exp\left\{ \frac{h}{4} \int_0^t \frac{g''(s)}{g(s)}\, \mathrm{d}s + O(h^2) \right\}. \tag{16}$$

Because $g''/g > 0$, we get $E_c = 0$ and $E_f = \Theta(h)$. In odd dimension the sum has the order of the lower bound in Theorem 3. Mix the exact coefficient and the secant coefficient with a weight θ: $(1-\theta)g'/g + \theta[g(t+h) - g(t)]/[hg(t)]$. The leading coarse and refined error coefficients are then proportional to $|\theta - 1|$ and $|\theta - 1/2|$. Their sum is constant for $1/2 \le \theta \le 1$. The order does not change.

A. The Smallest Leading Error in One Dimension

Let $d = T = 1$ and $\sigma = 1$. Use uniform grids of N steps and divide each step into m equal parts. Let $q_N^{(m)}$ be the infimum of $E_c + E_r$ over all fields in $\mathcal{B}_K$, not only linear fields. The exact generator has the coefficient $c(t) = g'(t)/g(t)$. If K is larger than this coefficient and its first two derivatives, then

$$\lim_{N\to\infty} N q_N^{(m)} = \tfrac{1}{2}(1 - 1/m)(1 + \pi/2). \tag{17}$$

For bisection the constant is $(2+\pi)/8$. The secant field attains it in the limit. The proof has three steps. In one dimension the exact generator of this path is unique. For small steps the Euler maps are increasing, so the pairing of equal quantiles attains W_2. An expansion of the particle error that is uniform over the class then gives the lower bound (Supplementary Theorem S10).

B. Smoothness in Time and Intermediate Times Are Necessary

A field that oscillates in time can be exact on a grid and on its bisection at each coarse node. Its first derivatives are bounded, but its second time derivative increases like N. The bound on $\partial_{tt} u$ in (1) is therefore necessary (Supplementary Section S8).

An error at the final time alone does not give the lower bound. The field $u(t,x) = -5t^2 x$ on $[0,1]$ generates a Gaussian path. Euler has a second-order error at the final time for each fixed refinement factor. Its error over the full path is first order. Acceleration effects of opposite sign cancel at the final time (Supplementary Section S11).

[7] AI assistance for this section is disclosed in the acknowledgment [6–9].

VIII. Numerical Experiments

We first compute the errors for fields with known distribution paths. Then we measure what happens when networks are trained for a small number of steps. In one dimension we estimate population errors by numerical integration. In two dimensions we examine how much the comparison changes with the evaluation samples. Supplementary Sections S14–S16 give all settings.[8]

A. Exact Population Computations

Fig. 1(b) shows the scalar Gaussian path of Section VII. For the secant field, $N(E_c + E_f)$ is 0.666 at $N = 8$ and 0.6428 at $N = 4096$. The limit in (17) is $(2+\pi)/8 \approx 0.6427$. The exact generator has the limit $3(2+\pi)/8$.

Fig. 1(c) shows the two other regimes. For first order, take the path (9) in dimension three. The first target coordinate is an equal mixture of Gaussians with means ± 1.5 and standard deviation 0.5. The other two coordinates are centered Gaussians with the same standard deviation. We compare two exact generators. The first is the conditional field. The second uses the skew field of Section V in the Gaussian plane, which makes Euler exact in that plane. Theorem 3 gives a lower bound c/N for both. With the conditional field, $E_c + E_f$ decreases from 0.292 at $N = 8$ to 0.0096 at $N = 256$. With the skew field, it decreases from 0.228 to 0.0074. The observed orders between $N = 128$ and $N = 256$ are 0.997 and 0.998. The skew field removes the error in the plane, and the error in the mixture coordinate stays at first order.

For second order, take $d = 2$ and the rotation field $u(t,x) = \tanh(t)x + Jx$, where J is the rotation by a quarter turn. It generates $p_t = \mathcal{N}(0, \cosh^2(t) I)$. Its acceleration $2\tanh(t)Jx$ moves particles around circles. Rotation leaves the Gaussian unchanged, so the acceleration preserves each p_t. Theorem 1 predicts second order. The scaled error $N^2(E_c + E_f)$ is 0.68 at $N = 8$ and 0.79 at $N = 256$.

B. Learned Scalar Fields With Population Errors

In one dimension, W_2 pairs points with the same cumulative probability. For example, it pairs the two medians. We estimate population errors by quadrature, which approximates an integral with a weighted sum. The target is the mixture coordinate above with the path (9), so Theorem 3 applies with $d = 1$.

We train networks with two hidden layers of 48 tanh units on three objectives, with ten random seeds per objective. *CFM* uses the conditional flow matching loss [1]. The *one-grid fit* adds a loss for agreement with the target path in a run of eight Euler steps. The loss uses the times $1/4$, $1/2$, $3/4$, and 1. The *two-grid fit* adds this loss for runs of eight and of sixteen steps. The added loss is the squared W_2 distance between sorted batches. The step size is not an input of the networks. Training uses Adam [32], and all settings were fixed before the evaluation.

We report three quantities. The *target error* at N steps is the largest W_2 distance to the target path at the four times. The *integration error* is the largest root mean squared (RMS)

[8] AI assistance for this section is disclosed in the acknowledgment [6–9].

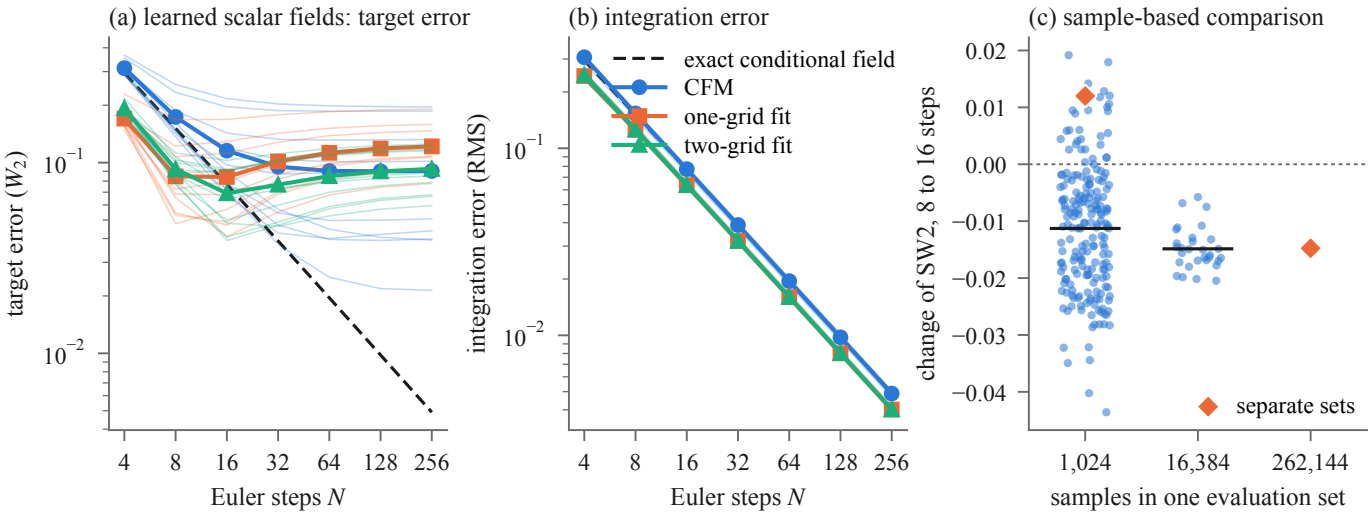


Fig. 2. Learned fields. (a) Target error of learned scalar fields in the population W_2 distance. Thin lines are the ten seeds of each objective, and lines with markers are their means. Fields that are fitted at 8 steps, or at 8 and 16 steps, have their smallest target error near those step counts. (b) The integration error of the same fields decreases at first order for all objectives. The legend is for panels (a) and (b). (c) Change of the empirical sliced Wasserstein distance (SW2) from 8 to 16 steps, averaged over six fitted two-dimensional fields. Positive values show increases. Each dot is one evaluation set. The lines give the means for each set size, and the diamonds give the values for the separate sets of 1,024 and 262,144 samples.

TABLE I
LEARNED SCALAR FIELDS: MEANS OVER TEN SEEDS OF POPULATION ERRORS AT N EULER STEPS. THE EXACT FIELD IS THE CONDITIONAL VELOCITY, WITHOUT TRAINING. SUPPLEMENTARY TABLE S7 GIVES EACH FIELD.

	Target error				Model	Integration error	
Field	N=8	16	64	256	bias	N=8	256
Exact field	0.151	0.077	0.020	0.005	0.000	0.151	0.005
CFM	0.174	0.115	0.090	0.090	0.091	0.153	0.005
One-grid fit	0.085	0.084	0.113	0.122	0.125	0.125	0.004
Two-grid fit	0.092	0.069	0.085	0.093	0.096	0.126	0.004

distance to the exact flow of the field from the same start points. The *model bias* is estimated as the largest W_2 distance between a reference solution of the flow and the target at 64 times. The errors E_c and E_f of (2) use all nodes of the coarse grid. Supplementary Table S7 gives them for $N = 8$.

Table I and Fig. 2(a,b) give the results. The integration error decreases at about first order for all objectives. The target error does not. For the CFM fields, it decreases from 0.174 at 8 steps to 0.090 at 256 steps. Their mean model bias is 0.091, with values between 0.023 and 0.195 for the ten seeds. The one-grid fit decreases the mean target error at 8 steps to 0.085. For these fields the target error is larger at 64 steps and at 256 steps than at 8 steps, for all ten seeds. The exact two-sided Wilcoxon signed-rank test gives $p = 0.002$ for each of the two comparisons. The change from 8 to 16 steps has no consistent direction: the target error increases for 6 of the ten seeds ($p = 0.695$). Between 8 and 256 steps the mean target error increases by a factor of 1.4, while the mean integration error decreases from 0.125 to 0.004. For the two-grid fields, the target error is smallest at 16 steps for 9 of the ten seeds. For all 20 fitted fields, the target error is larger at 64 steps and at 256 steps than at 16 steps. The test gives $p = 0.002$ for each objective and each comparison. For all ten CFM fields, the target error is smaller at 64 and at 256 steps than at 16 steps.

For a fixed field, the target error at the four evaluation times approaches the bias of the exact flow at those times. A coarse run can offset part of this bias. In the fitted fields, finer runs approach the exact flow as their error against the target increases. These comparisons hold the field fixed. Theorem 1 instead concerns the best error over a class of fields.

C. Sample-Based Comparisons in Two and Eight Dimensions

In higher dimension, the target error is estimated from samples. We test how reliable a comparison of two step counts then is. We train 18 tanh networks on mixtures of four Gaussians in dimensions two and eight, with three seeds for each of the objectives CFM, one-grid fit, and two-grid fit. Here the added losses compare averages of random Fourier features [33], and the two-grid fit also has a term for agreement between the two runs (Supplementary Section S14). We project the samples onto a line in each of 128 fixed directions and compute the scalar W_2. The root mean square over directions is the empirical sliced Wasserstein distance (SW2) [34]. We report its maximum over five times. We call the one-grid and two-grid fields the *fitted fields*.

On the 1,024-sample evaluation set, all six fitted two-dimensional fields have a larger SW2 at 16 steps than at 8 steps. On the independent set of 262,144 samples, all nine fields have a smaller SW2 at 16 steps (Supplementary Tables S1 and S5). To measure how often the direction changes, we evaluated the same fields on 230 sets of two sizes (Fig. 2(c), Supplementary Section S16). Take first the sets of 1,024 samples. The change for a fitted field has a standard deviation of 0.014 between sets and a mean of -0.011. These values are averages over the six fields. SW2 increases in 6% to 38% of the 200 sets, as a function of the field. All six fields show an increase together in 2.5% of the sets, as they do in the separate 1,024-sample

set. With 16,384 samples, SW2 increases in at most 7% of the 30 sets per field. On the 262,144-sample set, it decreases for all six. A single small set can give the opposite direction to the larger set for these fields. For the CFM fields, SW2 decreases from 8 to 16 steps in all sets.

The change from 8 to 64 steps differs between the fields. With 16,384 samples, SW2 is larger at 64 steps than at 8 steps in 100%, 30%, and 77% of the sets for the three one-grid fields. The values for the three two-grid fields are 33%, 7%, and 20%. For the one-grid field with the value 100%, simultaneous confidence intervals for a finite set of bounded test functions [35–37] also establish the increase for that set of functions (Supplementary Sections S12 and S14-E). The 8-to-64 increase therefore depends on the field and the measurement in this two-dimensional example.

IX. Relation to Prior Work

Fields that generate one path. The division of a vector field into a part that changes a distribution and a part that preserves it is classical [21, 25]. Obzherin and Khrulkov use it for Euler sampling. For a field that generates the path, the leading one-step error in W_2 is $h^2/2$ times the norm of the projected acceleration [22, Theorem 2]. Small projected coefficients are sufficient for a higher order [23, Corollary 8]. Table II compares their results with ours. The difference is the direction of the inference and the bias of the field. We assume accuracy of two Euler runs and we conclude a property of a limit field. The subtraction of the two runs removes the unknown model bias. Discretization bias as a cause of good samples is studied in [38].[9]

Solvers, schedules, and paths. Bespoke solvers learn a consistent update for a fixed field [13]. Other work chooses time steps from estimates of how quickly the velocity changes [39]. Expansions of the error of test averages are available for diffusion schedules [40], and a Gaussian path can be separated from its training schedule [41]. Our comparison constrains one unchanged field in each pair of runs, also when its continuous distribution is biased.

Straight flows. Straight trajectories make Euler exact, and optimal transport gives such trajectories for a different coupling [14, 15, 42, 43]. For linear interpolation of a given coupling, the conditional velocity has straight trajectories only if the coupling is deterministic [44, Theorem 2.6]. Theorem 3 keeps the distributions of the independent coupling. In odd dimension it rules out a generator in the allowed class whose acceleration preserves each distribution.

Error bounds for a given velocity. Upper bounds on the sampling error of flow matching and of the probability flow ODE include the error of the learned field [45–49]. Tao and Choi [50] relate the integration error to spatial derivatives of the velocity. Those results bound the error of one field from above. Our result is a necessary condition on all fields in a class.

Models with a step-size input. Shortcut models [19] take the step size as an input, and MeanFlow models [20] learn an average velocity. One set of parameters with two different step-size inputs is *not* one field in the sense of Theorem 1. The theorem therefore does not limit those methods.

X. Conclusion

The target error of a numerical run depends on the solver and on the learned field. For Euler on a grid and on its bisection, the difference of the two runs removes the unknown model bias from the leading error terms. The best two-grid error can decay faster than first order exactly when a limit of the allowed fields generates the target path and its acceleration preserves each distribution. For the independent interpolation path in odd dimension, no field in the class has this property. The experiments show the practical consequence. A field that is tuned for one step count can have a larger target error at larger step counts. Reports of sampling accuracy should therefore give the target error and the integration error separately. They should also give the step counts, the times, and the evaluation set.[10]

Limitations. The theorems use population Wasserstein errors at shared times along a prescribed path. They need global derivative bounds, a Gaussian start, and deterministic grids that all particles share. The lower-bound constant in Theorem 1 comes from a compactness argument and has no formula. The explicit constants of Theorems 2 and 4 are too large for numerical use: the certified derivative bounds of the 30 networks of Supplementary Section S14 are between 12,096 and 50,688. The sharp constant is for one scalar path on uniform grids. The existence of an order-elevating generator for the independent interpolation path in even dimension is open when the target is not an isotropic Gaussian. The learned-field experiments use scalar and low-dimensional mixtures. They show the offset of numerical error and model bias for single fields, and they do not test the rate of Theorem 1 over a class.

Future work. Four problems are open: order-elevating generators for targets that are not radial, sharp constants in higher dimension, weaker assumptions on smoothness in time, and conditions under which a small residual implies a small distance to an exact generator. Three cases need a separate analysis: fields with a step-size input, grids that depend on the particle, and objectives that use only the final time.

Data and Code Availability

The supplementary archive contains the code, the settings, the saved results, and the coefficients of all fitted networks. Scripts in the archive recompute each table and figure of this article.

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[9] AI assistance for this section is disclosed in the acknowledgment [6–9].

[10] AI assistance for this section is disclosed in the acknowledgment [6–9].

TABLE II
ASSUMPTIONS AND CONCLUSIONS OF THE CLOSEST PRIOR RESULTS.

Question	Prior result	Result here
Leading Euler error [22, Thm. 2], [23, Cor. 8]	The field generates the target path. The projected acceleration gives the leading one-step error. Small projected coefficients are sufficient for a higher final-time order on a flat torus.	If the two-grid error decays faster than first order, the closure contains an exact generator whose acceleration preserves each distribution. The fitted fields can be biased (Theorem 1).
Measured residual [23, Prop. 10]	A finite Gram matrix separates the measured residual from an orthogonal remainder.	Two target errors bound the measured coefficients. Gaussian regularity bounds the remainder (Theorem 2).
Odd dimension [22, Thm. 12], [23, Thm. 34]	For isotropic Gaussian paths, no linear field removes the leading Euler error in odd dimension. For one Euler step, a tangential field that maps each sphere to a larger sphere needs even dimension.	In odd dimension no field in $\overline{\mathcal{B}}_K$, linear or nonlinear, is an order-elevating generator of the independent interpolation path, for centered targets with positive definite covariance. The lower bound holds for changing fields and grids (Theorem 3).
Initial loss [23, Prop. 30]	For a field that generates the path, a second-moment defect of the correction energy is a lower bound on the projected acceleration.	Two numerical target errors constrain the initial loss, without exact generation (Theorem 4).

5.5 assisted with review of the claims and their presentation [8]. Anthropic Claude Fable 5.2 assisted with writing, mathematical analysis, numerical verification, reference checks, and the experiments of Sections VIII-B and VIII-C [9]. The author takes responsibility for the scientific content and attribution.

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